

Time Series Analysis (EN.553.439/639)

Full Course Design Package

Reverse-engineered from the Spring 2026 syllabus (Instructor: Dr. Sergey Kushnarev)

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Contents

1	Course Overview and Design Assumptions	1
2	Module 1: Foundations — Description, Stationarity, Trends & Linear Models	2
2.1	Topic 1: Fundamental Tools — Mean, Variance, Covariance, ACF	2
2.2	Topic 2: Stationarity & Basic Models	3
2.3	Topic 3: Trends	4
2.4	Topic 4: Linear Models — General Linear Process, AR, MA, ARMA	4
3	Module 2: Nonstationarity, Identification, Estimation & Diagnostics	5
3.1	Topic 5: Nonstationary Models — Differencing, ARIMA, Transformations	5
3.2	Topic 6: Model Specification — ACF, PACF, EACF	6
3.3	Topic 7: Parameter Estimation — Yule–Walker, MLE, LS/CLS	7
3.4	Topic 8: Model Diagnostics	7
4	Module 3: Forecasting, Seasonality & Time Series Regression	8
4.1	Topic 9: Forecasting	8
4.2	Topic 10: Seasonal Box–Jenkins Models — SARIMA	9
4.3	Topic 11: Regression Methods — Intervention Analysis, Outliers, Spurious Correlation, Pre-whitening	10
5	Module 4: Advanced & Special Topics (Enrichment)	10
5.1	Topic 12: Heteroskedastic Models — ARCH/GARCH	11
5.2	Topic 13: Nonlinear Time Series & Spectral/Frequency-Domain Methods	11
6	Prerequisite Map	12
7	Assessment Plan	13

1 Course Overview and Design Assumptions

This design package reverse-engineers the EN.553.439/639 syllabus into four instructional modules, a topic-by-topic objective map, a prerequisite dependency table, a difficulty analysis, and an assessment plan. The following assumptions were required to fill gaps the syllabus itself leaves open:

- **Duration.** The brief specified 12 weeks, but the syllabus’s own midterm dates (Feb. 18 = “Week 5” Wed, Mar. 25 = “Week 9” Wed, Apr. 22 = “Week 13” Wed) are only consistent with a **13-teaching-week term plus a non-teaching spring-break week** between Weeks 8 and 9. This document follows the syllabus’s own week numbering rather than compressing it.
- **Chapter mapping.** The syllabus states topics cover “approximately the first ten chapters and parts of Chapter 12” of Cryer & Chan (2nd ed.) without naming exact chapter numbers per topic; the mapping used here follows that text’s standard chapter order.
- **Pacing.** No lecture-by-lecture schedule was available (the syllabus defers this to CANVAS); week ranges below are inferred from topic count, the three midterm boundaries, and typical pacing for this material.
- **No final exam.** Grading sums to 100% via 6 homeworks (25%) plus 3 non-cumulative midterms (75%). Topics 12–13 (ARCH/GARCH; nonlinear & spectral methods) are exactly the topics the syllabus itself calls “other topics as time allows,” so they are treated here as ungraded enrichment rather than examined material.
- **Prerequisite gaps flagged, not assumed away.** OLS regression (needed for Trends) is not explicitly named among the stated prerequisite courses; Fourier/complex-number background (needed for spectral analysis) is not in the prerequisites at all. Both are flagged in the Prerequisite Map.
- **Undergraduate/graduate sections.** EN.553.439 and EN.553.639 share lectures; this design targets the shared “intermediate” level the syllabus itself uses, without inventing an undocumented graduate-only track.

2 Module 1: Foundations — Description, Stationarity, Trends & Linear Models

Weeks 1–5 (Midterm 1: Wednesday, Week 5). Goal: equip students to characterize a time series descriptively and represent it as a stationary ARMA process.

2.1 Topic 1: Fundamental Tools — Mean, Variance, Covariance, ACF

Core Concepts:

- Time series as a realization of a stochastic process $\{Y_t\}$
- Mean function $\mu_t = E[Y_t]$
- Autocovariance function $\gamma(t, s) = \text{Cov}(Y_t, Y_s)$
- Autocorrelation function (ACF) $\rho_k = \gamma_k/\gamma_0$
- Sample ACF and its approximate $\pm 2/\sqrt{n}$ significance bounds
- Time series plots and lag plots for exploratory analysis

Learning Objectives. Students will be able to:

1. *Define* the mean, autocovariance, and autocorrelation functions of a stochastic process.

2. *Compute* the sample ACF for a dataset and *interpret* a correlogram.
3. *Explain* why time-ordering and dependence invalidate classical iid-based inference.
4. *Distinguish* the population ACF from its sample estimate, including sampling variability.

Example Questions:

1. **(Recall)** Define the autocorrelation function ρ_k of a stationary process in terms of its autocovariance function.
Answer: $\rho_k = \gamma_k / \gamma_0$, where $\gamma_k = \text{Cov}(Y_t, Y_{t+k})$ and $\gamma_0 = \text{Var}(Y_t)$.
2. **(Application/Analysis)** Given sample ACF values $r_1 = 0.62$, $r_2 = 0.35$, $r_3 = 0.10$, $r_4 = -0.05$ for $n = 100$, use the $\pm 2/\sqrt{n}$ rule to determine which lags are significant.
Answer/Rubric: $2/\sqrt{100} = 0.2$; lags 1 and 2 exceed the threshold (significant), lags 3–4 do not. Full credit requires the correct threshold *and* correct classification of each lag; partial credit for the threshold alone.

2.2 Topic 2: Stationarity & Basic Models

Core Concepts:

- Strict stationarity vs. weak (second-order) stationarity
- Conditions under which the two coincide (e.g., Gaussian processes)
- White noise process
- Random walk (with/without drift) as a canonical nonstationary process
- MA(1) as a first stationary example

Learning Objectives. Students will be able to:

1. *Define* strict and weak stationarity and state the conditions distinguishing them.
2. *Determine* whether a given process is stationary by checking its mean/variance/covariance structure.
3. *Derive* the ACF of a simple process (MA(1), random walk) from first principles.
4. *Justify* why stationarity is a prerequisite for meaningful ACF-based inference.

Difficulty Note. *Why hard:* students trained on iid statistics have never needed a joint-distributional property to be time-invariant, and commonly conflate “stationary” with “looks flat,” so they can recite the definition but cannot apply it to a concrete process. *Mitigation:* before the formal definition, show white noise vs. a same-starting-value random walk side by side (one wanders, one does not, yet neither plot alone proves anything), then walk line-by-line through a 3-process checklist (white noise, random walk, MA(1)) asking “does this expression depend on t ?”

Example Questions:

1. **(Recall)** State the definition of weak (second-order) stationarity.
Answer: Constant mean $E[Y_t] = \mu$ for all t ; constant finite variance; $\text{Cov}(Y_t, Y_{t+k})$ depends only on lag k , not on t .

2. **(Application)** Is the random walk $Y_t = Y_{t-1} + e_t$, $Y_0 = 0$, stationary? Justify using its variance function.

Answer/Rubric: $\text{Var}(Y_t) = t\sigma_e^2$, which depends on t , so Y_t is not stationary. Full credit requires the explicit variance derivation, not just “no.”

2.3 Topic 3: Trends

Core Concepts:

- Deterministic trend models (linear, quadratic, seasonal-means regression)
- Estimating and removing deterministic trends via least squares
- Residual analysis after detrending
- Stochastic (unit-root) trend as an alternative source of nonstationarity
- Overdifferencing vs. overfitting a deterministic trend

Learning Objectives. Students will be able to:

1. *Fit* a deterministic trend model via least-squares regression.
2. *Distinguish* deterministic trend behavior from stochastic trend behavior in a plot and its ACF.
3. *Evaluate* whether detrended residuals are consistent with a stationary process.
4. *Critique* the consequences of misclassifying a stochastic trend as deterministic (or vice versa).

Example Questions:

1. **(Recall)** What distinguishes a deterministic trend from a stochastic trend?

Answer: A deterministic trend is a fixed function of time (e.g., $\beta_0 + \beta_1 t$) with stationary deviations around it; a stochastic trend is nonstationarity built into the process itself (e.g., a unit root), so shocks have a permanent effect on the level.

2. **(Application/Analysis)** A researcher fits $Y_t = \beta_0 + \beta_1 t + e_t$ by OLS and finds the residual ACF decays very slowly. What does this suggest, and what would you try next?

Answer/Rubric: Suggests the series may have a stochastic trend (or the deterministic form is misspecified) rather than stationary deviations; next step is to difference the original series and re-examine the ACF. Full credit requires both the diagnosis and a concrete next step.

2.4 Topic 4: Linear Models — General Linear Process, AR, MA, ARMA

Core Concepts:

- General linear process $Y_t = \sum_j \psi_j e_{t-j}$
- Backshift operator B ; operator polynomials $\phi(B)$, $\theta(B)$
- $\text{AR}(p)$, $\text{MA}(q)$, $\text{ARMA}(p, q)$ definitions
- Stationarity condition (roots of $\phi(z) = 0$ outside unit circle) and invertibility condition (roots of $\theta(z) = 0$ outside unit circle)

- Theoretical ACF and PACF shapes for AR/MA/ARMA processes

Learning Objectives. Students will be able to:

1. *Express* an AR, MA, or ARMA process using backshift-operator notation.
2. *Derive* the stationarity condition for an $AR(p)$ process from the roots of its characteristic polynomial.
3. *Derive* the theoretical ACF of an $AR(1)/AR(2)$ or $MA(1)/MA(2)$ process.
4. *Evaluate* whether given estimated coefficients yield a stationary and invertible model.

Difficulty Note. *Why hard:* this is the first place students must connect a difference equation to a polynomial-root condition, requiring fluency with B as an algebraic object rather than a time-shift description; $\phi(B)$ and the characteristic equation $\phi(z) = 0$ look alike and are frequently conflated. *Mitigation:* hand-iterate an $AR(1)$ recursion numerically for $\phi = 0.5$ and $\phi = 1.5$ so students see one path settle and one explode *before* the root condition is introduced as “the algebra that predicts what you just watched.”

Example Questions:

1. **(Recall)** Write the $AR(2)$ model in backshift-operator form and state its characteristic equation.
Answer: $(1 - \phi_1 B - \phi_2 B^2)Y_t = e_t$; characteristic equation $1 - \phi_1 z - \phi_2 z^2 = 0$.
2. **(Application/Analysis)** For $Y_t = \phi Y_{t-1} + e_t$, find the range of ϕ for stationarity and compute ρ_k .
Answer/Rubric: Stationary for $|\phi| < 1$; $\rho_k = \phi^{|k|}$. Credit for correct range *and* correct ACF formula; partial credit if only one is correct.

3 Module 2: Nonstationarity, Identification, Estimation & Diagnostics

Weeks 6–9 (spring break falls between Weeks 8 and 9; Midterm 2: Wednesday, Week 9). Goal: run the full Box–Jenkins loop — difference/transform, identify, estimate, diagnose — on real, nonstationary data.

3.1 Topic 5: Nonstationary Models — Differencing, ARIMA, Transformations

Core Concepts:

- Differencing operator $\nabla = 1 - B$; order- d differencing
- $ARIMA(p, d, q)$ as ARMA applied to a differenced series
- Variance-stabilizing transformations (log, Box–Cox)
- Random walk as $ARIMA(0, 1, 0)$; overdifferencing pitfalls

Learning Objectives. Students will be able to:

1. *Apply* differencing to transform a nonstationary series into an apparently stationary one.
2. *Select* an appropriate variance-stabilizing transformation from a plot showing changing variance.

3. *Formulate* a series' dynamics as an $\text{ARIMA}(p, d, q)$ model.
4. *Assess* evidence of overdifferencing.

Example Questions:

1. **(Recall)** What operation converts an $\text{ARIMA}(p, d, q)$ process into a stationary $\text{ARMA}(p, q)$ process?
Answer: Differencing d times: $(1 - B)^d Y_t$.
2. **(Application)** A series shows increasing variance over time and a slowly decaying ACF. Propose a sequence of steps to reach a stationary series, and justify each.
Answer/Rubric: (1) Apply a variance-stabilizing transform such as log (justified by increasing variance); (2) then difference the transformed series (justified by slow ACF decay); (3) re-check the ACF/plot. Full credit requires the correct order (transform before differencing) and justification for each step.

3.2 Topic 6: Model Specification — ACF, PACF, EACF

Core Concepts:

- Partial autocorrelation function (PACF) via Durbin–Levinson/regression definition
- $\text{AR}(p)$: PACF cuts off at lag p , ACF tails off; $\text{MA}(q)$: ACF cuts off at lag q , PACF tails off; $\text{ARMA}(p, q)$: both tail off
- Extended ACF (EACF) triangular pattern
- Using ACF/PACF/EACF jointly with parsimony to propose candidate models

Learning Objectives. Students will be able to:

1. *Compute* and *interpret* the sample PACF of a time series.
2. *Identify* candidate (p, q) orders from ACF/PACF cutoff/tailoff patterns.
3. *Read* an EACF table to shortlist candidate ARMA orders.
4. *Compare* competing candidate specifications and justify a preferred one.

Difficulty Note. *Why hard:* identification is pattern recognition against a lookup table, but real sample ACF/PACF plots are noisy, so a genuine $\text{AR}(2)$ may not cut off crisply at lag 2; the EACF adds a second, unfamiliar 2-D scanning task (finding a triangle's vertex) unlike the 1-D “read bar k ” of an ACF. *Mitigation:* pair a clean textbook ACF/PACF with a messy real one on the same day and practice ranking several plausible candidates rather than identify-and-stop; for EACF, walk through a table with the vertex pre-highlighted before asking students to find one unaided.

Example Questions:

1. **(Recall)** What are the theoretical ACF and PACF signatures of an $\text{MA}(2)$ process?
Answer: ACF cuts off (equals 0) after lag 2; PACF tails off (decays), never exactly zero.
2. **(Application/Analysis)** A sample ACF tails off gradually and a sample PACF cuts off sharply after lag 1. What model would you tentatively identify, and why?
Answer/Rubric: $\text{AR}(1)$ — a PACF cutoff at lag 1 combined with a tailing-off ACF is the $\text{AR}(1)$ signature. Must name $\text{AR}(1)$ specifically and cite both patterns.

3.3 Topic 7: Parameter Estimation — Yule–Walker, MLE, LS/CLS

Core Concepts:

- Method-of-moments/Yule–Walker equations for AR parameters
- Conditional and unconditional least squares (CLS/ULS) for ARMA
- Maximum likelihood estimation (exact vs. conditional likelihood)
- Asymptotic properties: consistency, approximate normality, standard errors

Learning Objectives. Students will be able to:

1. *Derive* the Yule–Walker equations for an $AR(p)$ process and solve them for $AR(1)/AR(2)$.
2. *Contrast* CLS, ULS, and MLE in terms of assumptions and computational approach.
3. *Compute* parameter estimates and standard errors for a fitted ARMA model in software.
4. *Evaluate* which estimation method is most appropriate given sample size and complexity.

Difficulty Note. *Why hard:* students must hold three estimation methods in mind and often can’t articulate what differs beyond “one uses calculus”; this is worsened because exact MLE is usually taught via a conditional-likelihood shortcut without covering why the exact likelihood is hard (the innovations-algorithm prerequisite is typically missing). *Mitigation:* estimate the same $AR(1)$ example three ways in one side-by-side table so students see the methods converge rather than treating them as unrelated procedures.

Example Questions:

1. **(Recall)** Write the Yule–Walker estimator of ϕ for an $AR(1)$ process.
Answer: $\hat{\phi} = r_1$ (the lag-1 sample autocorrelation), from $\rho_1 = \phi$ for $AR(1)$.
2. **(Application/Analysis)** Explain one key difference between CLS and MLE for ARMA estimation, and one condition under which they should agree closely.
Answer/Rubric: CLS conditions on (fixes) initial values, ignoring their contribution to the likelihood; exact MLE incorporates the full joint density including initial observations. They converge for large n , since the initial conditions’ contribution becomes negligible. Credit for a correct distinguishing feature and a correct convergence condition.

3.4 Topic 8: Model Diagnostics

Core Concepts:

- Standardized residuals; residual ACF
- Ljung–Box (portmanteau) test for residual autocorrelation
- Normality checks (QQ-plot, Shapiro–Wilk) and overfitting checks
- The iterative identify $\rightarrow$ estimate $\rightarrow$ diagnose ($\rightarrow$ revise) loop

Learning Objectives. Students will be able to:

1. *Perform* residual diagnostics (residual ACF, Ljung–Box) on a fitted ARIMA model.

2. *Interpret* the null hypothesis and result of a Ljung–Box test.
3. *Diagnose* evidence of misspecification and propose a revised model.
4. *Justify* when a model should be judged adequate versus requiring refinement.

Example Questions:

1. **(Recall)** What null hypothesis does the Ljung–Box test check when applied to fitted-model residuals?
Answer: H_0 : the residuals are uncorrelated up to the tested lag (the model has captured the linear dependence; residuals resemble white noise).
2. **(Application/Analysis)** A fitted model’s residual ACF shows a significant spike at lag 12, with all other lags within bounds. What would you investigate, and what change might you consider?
Answer/Rubric: Investigate seasonal dependence (period 12, e.g. monthly data); consider adding a seasonal AR or MA term (moving to a SARIMA specification). Credit for identifying seasonality and proposing a seasonal remedy.

4 Module 3: Forecasting, Seasonality & Time Series Regression

Weeks 10–13 (Midterm 3: Wednesday, Week 13). Goal: use fitted models to forecast, extend ARIMA to seasonal data, and incorporate external events/outliers via regression.

4.1 Topic 9: Forecasting

Core Concepts:

- Minimum mean-square-error (MMSE) forecast = conditional expectation
- ψ -weight forecasting and forecast-error variance
- Forecast recursions for AR, MA, ARMA, ARIMA
- Prediction intervals and their widening with forecast horizon

Learning Objectives. Students will be able to:

1. *Derive* the h -step-ahead MMSE forecast for an AR(1) or MA(1) process.
2. *Compute* forecasts and standard errors for a fitted ARIMA model using ψ -weights.
3. *Construct* a prediction interval at a specified confidence level.
4. *Explain* why prediction-interval width grows unboundedly with horizon for ARIMA ($d \geq 1$) processes.

Difficulty Note. *Why hard:* deriving forecast-error variance via ψ -weights is the heaviest symbolic-manipulation task in the course (infinite-series conditional expectations), and clashes with the intuition that a good model should forecast with growing certainty. *Mitigation:* build the derivation two terms at a time (1-step, then 2-step) with actual numbers before generalizing; pair the ARIMA($d = 1$) case with a visual contrasting a random-walk forecast fan (widens forever) against an AR(1) forecast (levels off).

Example Questions:

1. **(Recall)** For an MA(1) model, what is the 2-step-ahead forecast given only current information?
Answer: The unconditional mean (0 if mean-zero) — MA(1) has memory of only 1 lag, so nothing from the current information set informs step 2.
2. **(Application/Analysis)** Explain why the 95% prediction interval for an ARIMA(0,1,0) forecast widens without bound as horizon grows, while a stationary AR(1)'s levels off.
Answer/Rubric: Random-walk forecast-error variance equals $h\sigma_e^2$, growing linearly and unboundedly with h ; stationary AR(1) forecast-error variance converges to $\sigma_e^2/(1-\phi^2)$ as $h \rightarrow \infty$, giving a bounded interval. Credit requires citing the correct growth behavior for both cases.

4.2 Topic 10: Seasonal Box–Jenkins Models — SARIMA

Core Concepts:

- Seasonal differencing operator $\nabla_s = 1 - B^s$
- Multiplicative seasonal model $\text{SARIMA}(p, d, q) \times (P, D, Q)_s$
- Identifying seasonal orders from ACF/PACF spikes at seasonal lags
- Forecasting and diagnostics for seasonal models

Learning Objectives. Students will be able to:

1. *Formulate* a seasonal series as a $\text{SARIMA}(p, d, q) \times (P, D, Q)_s$ model.
2. *Identify* seasonal AR/MA orders from spikes at multiples of the seasonal period.
3. *Apply* seasonal differencing to remove seasonal nonstationarity.
4. *Evaluate* a fitted SARIMA model's adequacy via seasonal and non-seasonal diagnostics.

Difficulty Note. *Why hard:* the notation stacks two polynomials, two differencing operators, and a period into one symbol, so a model like $\text{ARIMA}(1, 1, 1) \times (1, 1, 1)_{12}$ is frequently mis-expanded. *Mitigation:* have students expand a small example, e.g. $(1, 0, 0) \times (1, 0, 0)_4$, fully by hand into an explicit lag polynomial before ever fitting one in software.

Example Questions:

1. **(Recall)** In $\text{SARIMA}(1, 1, 1) \times (1, 1, 1)_{12}$, what does the subscript 12 represent, and what does $D = 1$ indicate?
Answer: 12 is the seasonal period (e.g., monthly data with annual seasonality); $D = 1$ means one order of seasonal differencing, $(1 - B^{12})$, to remove seasonal nonstationarity.
2. **(Application/Analysis)** A monthly series' ACF shows large, slowly-decaying spikes at lags 12, 24, 36, plus typical AR(1)-like short-lag decay. What seasonal and non-seasonal orders would you propose?
Answer/Rubric: Seasonal differencing ($D = 1, s = 12$) to address the seasonal-lag pattern, plus a low-order AR term (e.g., $p = 1$) for the short-lag structure. Full credit for identifying seasonal differencing and a plausible non-seasonal order.

4.3 Topic 11: Regression Methods — Intervention Analysis, Outliers, Spurious Correlation, Pre-whitening

Core Concepts:

- Time series regression with autocorrelated errors
- Intervention analysis via step/pulse indicator variables
- Additive outliers (AO) vs. innovational outliers (IO)
- Spurious correlation between independent nonstationary series
- Pre-whitening for valid cross-correlation/transfer-function inference

Learning Objectives. Students will be able to:

1. *Explain* why standard OLS inference is invalid under autocorrelated regression errors.
2. *Model* a known intervention using indicator variables.
3. *Distinguish* additive from innovational outliers by their impact on the series.
4. *Demonstrate*, via pre-whitening, why an apparent correlation between two series may be spurious.

Difficulty Note. *Why hard:* a genuine conceptual trap, not just computational — two independent random walks routinely show “significant” correlation, directly contradicting trained intuition, and students often resist the result even after seeing it demonstrated. *Mitigation:* run a live simulation generating many independent random-walk pairs and plotting the distribution of their correlations (far wider than the iid benchmark), then introduce pre-whitening as the fix tied directly to the diagnosed cause.

Example Questions:

1. **(Recall)** Distinguish an additive outlier (AO) from an innovational outlier (IO).
Answer: An AO affects only a single observation directly (a one-time shock, not propagated through the model dynamics); an IO enters through the error term and propagates through the dynamics, affecting subsequent observations too.
2. **(Application/Analysis)** Two independent random-walk series have a sample correlation of 0.82. Is this evidence of a real relationship? What procedure would you use, and what do you expect it to show?
Answer/Rubric: No — this is classic spurious correlation. Recommend pre-whitening both series (fit an appropriate ARIMA model to each, then correlate the residuals); expect the pre-whitened cross-correlation to be near zero/insignificant. Full credit requires identifying spurious correlation, naming pre-whitening, and the expected outcome.

5 Module 4: Advanced & Special Topics (Enrichment)

Week 14, ungraded. Goal: preview two extensions beyond linear, constant-variance ARMA theory, consistent with the syllabus’s own framing of this material as time-permitting (“other topics as time allows”).

5.1 Topic 12: Heteroskedastic Models — ARCH/GARCH

Core Concepts:

- Volatility clustering as a stylized fact of financial/economic series
- Conditional mean vs. conditional variance modeling
- $\text{ARCH}(q)$: variance as a function of past squared residuals
- $\text{GARCH}(p, q)$ as a parsimonious generalization
- Estimation (MLE) and diagnostics (Ljung–Box on squared residuals) for ARCH effects

Learning Objectives. Students will be able to:

1. *Explain* conditional heteroskedasticity versus ARMA’s constant-variance assumption.
2. *Test* a residual series for ARCH effects via Ljung–Box on squared residuals.
3. *Specify* an $\text{ARCH}(q)$ or $\text{GARCH}(p, q)$ model and interpret its parameters.
4. *Compare* ARCH and GARCH specifications for parsimony and fit.

Example Questions:

1. **(Recall)** What stylized fact motivates ARCH/GARCH models over standard ARMA models?
Answer: Volatility clustering — high-volatility periods tend to be followed by more high volatility, and calm periods by more calm, i.e., the conditional variance changes over time even when the mean structure looks like white noise.
2. **(Application/Analysis)** A fitted ARMA model’s residuals pass the Ljung–Box test, but the squared residuals show significant autocorrelation. What would you conclude, and what would you fit next?
Answer/Rubric: No linear mean-dependence remains, but conditional heteroskedasticity (ARCH effects) is present; fit an $\text{ARCH}(q)$ or $\text{GARCH}(p, q)$ model to the residual series. Credit requires distinguishing mean vs. variance dependence and proposing ARCH/GARCH.

5.2 Topic 13: Nonlinear Time Series & Spectral/Frequency-Domain Methods

Core Concepts:

- Limitations of linear (ARMA) models; brief survey of nonlinear alternatives
- Spectral density function as the Fourier transform of the ACVF
- Periodogram as an estimator of the spectral density
- Relating time-domain (ACF) and frequency-domain (spectral density) representations

Learning Objectives. Students will be able to:

1. *Explain* the relationship between the autocovariance function and the spectral density.
2. *Compute* and *interpret* a periodogram for a series with suspected periodic behavior.

3. *Identify* dominant frequencies/cycles from a periodogram.
4. *Discuss* when a nonlinear time series model may be preferable to a linear ARMA model.

Difficulty Note. *Why hard:* almost no student has a Fourier-analysis background, so switching from “indexed by time” to “indexed by frequency” is the largest representational jump in the course, arriving when bandwidth is already stretched. *Mitigation:* introduce the periodogram purely visually first — decompose a synthetic signal that is literally two sine waves plus noise, showing its unremarkable time plot next to its periodogram (two clean spikes) before any formula.

Example Questions:

1. **(Recall)** How is the spectral density $f(\omega)$ of a stationary process related to its autocovariance function γ_k ?
Answer: $f(\omega)$ is (proportional to) the Fourier transform of the ACVF: $f(\omega) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma_k e^{-ik\omega}$.
2. **(Application/Analysis)** A series’ periodogram shows a single sharp, dominant spike at frequency $\omega = \pi/6$, with all other frequencies near zero. What does this suggest, and what is the implied period?
Answer/Rubric: Suggests a strong periodic/cyclical component at that frequency; period = $2\pi/\omega = 2\pi/(\pi/6) = 12$ time units. Full credit requires the correct period calculation and correct interpretation of a periodogram spike as a dominant cycle.

6 Prerequisite Map

Topic	Prerequisite(s)	Where Taught
T1 Fundamental tools	Expectation, variance, covariance of random variables	Assumed background (probability prereq: 553.310/311/420/421/620)
T2 Stationarity/basic models	T1 (ACF/ACVF); joint distributions & independence	T1 in this course; joint distributions = probability prereq
T3 Trends	Simple/multiple OLS regression; T2 (stationary-residual concept)	Gap flagged: OLS not explicitly named in prereqs — assumed via linear algebra prereq (110.201/212, projections/least squares); T2 in this course
T4 ARMA/linear models	T1–T3; linear algebra (polynomial roots, matrix operations); infinite-series convergence	T1–T3 in this course; linear algebra = 110.201/212 prereq
T5 ARIMA/differencing	T4 (ARMA representation); difference-operator algebra; log/power transforms	T4 in this course
T6 ACF/PACF/EACF	T4 (theoretical AR/MA shapes), T5 (ARIMA); partial-correlation concept	T4–T5 in this course

Topic	Prerequisite(s)	Where Taught
T7 Estimation	T6 (identified order); likelihood/MLE theory; linear algebra (Toeplitz systems)	T6 in this course; light MLE background flagged
T8 Diagnostics	T7 (fitted residuals); hypothesis testing / χ^2 distribution	T7 in this course; hypothesis testing = probability prereq
T9 Forecasting	T4, T5, T7 (model representation & fit); conditional expectation	T4/T5/T7 in this course; conditional expectation = probability prereq
T10 SARIMA	T5, T6, T9 (ARIMA, identification, forecasting), extended seasonally	T5/T6/T9 in this course
T11 Regression/intervention	T3 (trend regression); T8 (residual diagnostics); T6 (cross-correlation for pre-whitening)	T3/T6/T8 in this course
T12 ARCH/GARCH	T4 (AR structure on squared residuals); T7 (MLE); conditional variance	T4/T7 in this course; conditional variance = probability prereq
T13 Spectral/nonlinear	T1 (ACVF, whose Fourier transform is the spectral density); complex numbers/Fourier series	T1 in this course; Fourier/complex background NOT in stated prereqs — gap flagged

7 Assessment Plan

Status	Assessment	Timing	Topics	Rationale
Official	HW1	End Wk2 (Fri)	T1(–T2)	Checks sample-ACF computation/interpretation before trend/ARMA content builds on it
Recommended	Ungraded in-class check	Wk2	T2	Surfaces the “stationary = flat” misconception before it compounds into T4’s stationarity-condition work
Official	HW2	End Wk4 (Fri)	T2–T4	Confirms detrending and ARMA representation right before Midterm 1
Official	Midterm 1	Wk5 Wed	T1–T4	Summative check on describe → stationarize → represent-as-ARMA while fresh
Recommended	In-class ACF/PACF exercise	Start Wk7	T6	Surfaces the clean-signature-vs-noisy-real-plot misconception before HW3 relies on correct reading
Official	HW3	End Wk7 (Fri)	T5–T6	Tests differencing/transform choice and model identification
Official	HW4	End Wk8 (Fri)	T7–T8	Tests estimation and diagnostics together, mirroring the real iterative loop

Status	Assessment	Timing	Topics	Rationale
Official	Midterm 2	Wk9 Wed	T5–T8	Non-cumulative check on the identify → estimate → diagnose cycle
Recommended	Spurious-correlation simulation/quiz	Start Wk12	T11	This misconception resists simple explanation — confront it before HW6 requires pre-whitening
Official	HW5	End Wk10/11 (Fri)	T9–T10	Tests forecasting computations and seasonal identification/fitting
Official	HW6	End Wk12 (Fri)	T11–T12	Final graded checkpoint, bundling regression/intervention with an intro ARCH/GARCH application
Official	Midterm 3	Wk13 Wed	T9–T11	Non-cumulative; T12–T13 intentionally excluded (syllabus’s “as time allows” framing)
Recommended	Ungraded concept-check discussion	Wk14	T12–T13	Gives closure/accountability for enrichment material without touching the fixed 25/75 grade split